CITSA Backend Club Activities Schedule

11 weeks, starting the weekend of Oct 10 | Saturdays: Concept | Sundays: Practical (with a framework)

Week	Saturday — Concept Session	Sunday — Practical Session (Framework)
Week 1	Foundations: the need for backend, backend ideology, and understanding HTTP for backend.	Set up a backend framework (e.g., Express.js, Django, or Spring Boot); build and run a basic HTTP server.
Week 2	Routing, and Serialization & Deserialization.	Build multiple routes with parameters; add endpoints that serialize/deserialize JSON data.
Week 3	Authentication & Authorization.	Implement login with JWT or session-based auth; protect a route with authorization checks.
Week 4	Validation & Transformations.	Add request validation to an endpoint and transform/sanitize incoming data.
Week 5	Handlers, Service & Repository pattern (Middlewares, request context).	Refactor a small app into handler/service/repository layers; write a custom middleware.
Week 6	Complete REST API design.	Design and build a full CRUD REST API for a sample resource, following REST conventions.
Week 7	Databases with Postgres.	Connect the framework to PostgreSQL; perform CRUD operations using an ORM or query builder.
Week 8	Caching, and Task Queues & Background Jobs.	Add a caching layer (e.g., Redis) to an endpoint; set up a task queue and run a background job.
Week 9	Full-text search (Elasticsearch), and Error Handling & Fault-Tolerant Systems.	Integrate Elasticsearch and build a search endpoint; add centralized error handling/retries.
Week 10	Production-grade configuration, and Logging, Monitoring & Observability.	Set up environment-based config; add structured logging and a monitoring tool.
Week 11	Backend Security, Scaling & Performance Engineering, and Concurrency & Parallelism.	Apply security best practices; load-test an endpoint and apply a scaling or concurrency technique.